

Social Network Analysis of Mannargudi Heritage Tourism: Structure, Centrality, and Circuit Optimization

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Abstract

This report maps the living social network of Mannargudi, consisting of people, places, rituals, crafts, festivals, food, nature assets, and digital/diaspora communities. Using Social Network Analysis (SNA), the network (24 nodes, 34 edges) is analysed to identify structural properties and optimize tourism circuit design. Key metrics show a sparse but structured network (density 0.123), small-world characteristics (average path length 2.66), and a strong central hub dominated by the Rajagopalaswamy Temple (highest betweenness centrality 1.570).

Introduction

Mannargudi is presented as a living cultural ecosystem anchored by the Rajagopalaswamy Temple, supported by artisan traditions, festivals, food systems, and surrounding nature assets. The network extends beyond the town through circuit temple towns and a global Tamil diaspora. The study models these interconnected components as a social network to understand relationships and tourism potential.

Problem Statement

Despite having rich cultural, economic, and ecological assets, Mannargudi's tourism potential is underutilized due to:

- Weak connections between key communities (artisan, diaspora, nature, digital, governance)
- Lack of structured tourism circuits integrating all assets
- Underdeveloped digital and institutional promotion
- Presence of structural holes limiting flow between important nodes

Methodology / Implementation

- Constructed a network of 24 nodes across 9 communities (Temple, Festival, Artisan, Food, Nature, Circuit, Digital, Diaspora, Governance)
- Established 34 edges representing real-world relationships (weighted: strong = 2, weak = 1)
- Applied standard SNA metrics:
 - Graph Density = 0.123
 - Average Path Length = 2.659
 - Diameter = 6
 - Clustering Coefficient = 0.398
 - Network Centralisation = 1.481

- Computed centrality measures (betweenness, closeness, clustering) for all nodes
- Identified key broker nodes and structural holes
- Analysed community-level connectivity and inter-community bridges

Results

- Central Hub: Rajagopalaswamy Temple dominates the network (BC = 1.5705, degree = 14)
 - Key Brokers:
 1. Bronze & Metal Craftsmen (0.3834)
 2. TN Tourism Board (0.3412)
 3. Sengamalam (0.2635)
 4. Big Bazaar Street Market (0.2253)
 - Network Properties:
 1. Sparse but structured (density 0.123)
 2. Small-world (avg. path $\approx$ 3 hops)
 3. Moderately clustered communities
 - Communities Identified: 9 distinct clusters with internal cohesion and external dependencies
 - Structural Holes Identified:
 1. Sengamalam $\leftrightarrow$ TN Tourism Board
 2. Artisan economy $\leftrightarrow$ Diaspora
 3. Nature assets $\leftrightarrow$ Temple network
 4. Festival calendar $\leftrightarrow$ Digital amplification
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Inferences

- Sengamalam as a high-impact node: Ranked #3 in betweenness, outperforming institutional nodes; strong untapped tourism leverage
- Festival system: Year-round demand exists, but lacks digital and institutional marketing
- Food network: Peripheral despite high tourism potential
- Circuit feasibility: Average path length (2.66) shows circuit towns already structurally connected
- Diaspora linkage: Weak but high-value connection; significant revenue potential if formalized

Conclusion

Mannargudi's network is a hub-dominated, small-world system centered on the Rajagopalaswamy Temple, with multiple underutilized bridges across communities. Closing key structural holes enables the formation of integrated tourism circuits (temple, nature, heritage), transforming Mannargudi from a short-duration visit into a multi-day destination. The town functions as the core of the Chola heritage network, with all surrounding circuits deriving from its concentrated cultural, ritual, and economic ecosystem.